

Math 50 Midterm 1: Drill Answer Key

Worked solutions to the Drills in Unit 1 and Unit 2

Read this first

Try each drill yourself before reading the solution. The drill numbers and wording follow the *current* course website (Unit 1 has Drills 1–22). Your saved PDF of the Unit 1 notes is an older snapshot with 17 drills, so its numbers differ and it lacks some drills (e.g. “Getting caught skipping class”, “Reverse-engineering a joint distribution”, “Mixing a Bernoulli and a Uniform”). Unit 2 drills have titles instead of numbers. The end-of-unit “Problems” (data/plotting tasks) are not included here. Every number was checked by exact arithmetic or simulation.

1 Unit 1 drills

Drill 1 — Sample spaces and events

For each situation, identify the random variable(s), the sample space, and give example events.

Solution.

- (a) R = the student’s answer. $S_R = \{\text{YES}, \text{NO}\}$ (or $\{1, 0\}$). Events: $\{\text{YES}\}$, $\{\text{NO}\}$, and the whole space.
- (b) N = the number of rolls needed to get the first 6. $S_N = \{1, 2, 3, \dots\}$ (infinite, since a 6 could take arbitrarily long). Events: $\{N = 1\}$, $\{N \leq 3\}$, $\{N \text{ is even}\}$.
- (c) (H, G) = height and gender. $S = [0, \infty) \times \{\text{M}, \text{F}\}$ (height is continuous; we could restrict to a plausible range like $[0, 250]$ cm). Events: $\{G = \text{F}\}$, $\{H > 180\}$, $\{H > 170, G = \text{M}\}$.

Drill 2 — Probabilities from a tree diagram

Two independent people, each supports a ballot measure with probability 0.6.

Solution. Tree: first person supports (0.6) or not (0.4); from each branch, second person supports (0.6) or not (0.4). Leaves (multiply along branches), writing (person 1, person 2):

$$(1, 1) : 0.6 \cdot 0.6 = 0.36, \quad (1, 0) : 0.6 \cdot 0.4 = 0.24, \quad (0, 1) : 0.4 \cdot 0.6 = 0.24, \quad (0, 0) : 0.4 \cdot 0.4 = 0.16.$$

These sum to 1. **(b)** Both: 0.36. **(c)** Exactly one: $(1, 0)$ or $(0, 1)$: $0.24 + 0.24 = 0.48$. **(d)** At least one = $1 - \mathbb{P}(\text{neither}) = 1 - 0.16 = 0.84$ (equivalently $0.36 + 0.48$).

Drill 3 — Sample spaces of compound random variables

Solution. (a) Heads: die +1 gives $\{2, \dots, 7\}$; tails: die +4 gives $\{5, \dots, 10\}$. So $S_Y = \{2, 3, \dots, 10\}$.
 (b) Red $\rightarrow Y \in \{0, 1\}$; green/blue $\rightarrow Y \in \{1, 2, 3, 4\}$. So $S_Y = \{0, 1, 2, 3, 4\}$.
 (c) Odd values give 1, 3, 5; even values give $2 \cdot \{2, 4, 6\} = 4, 8, 12$. So $S_Y = \{1, 3, 4, 5, 8, 12\}$.
 (d) No. Values 2, 3, 4 (heads, die 1, 2, 3) and 8, 9, 10 (tails, die 4, 5, 6) each arise one way, but 5, 6, 7 arise two ways. For example $Y = 5$ comes from (heads, die 4) and from (tails, die 1).

Drill 4 — Getting caught skipping class

$N = 35$ students, 5 checked uniformly at random each day; you skip half of all classes independently; $M = 27$ meetings.

Solution. (a) $\mathbb{P}(\text{checked}) = \frac{5}{35} = \frac{1}{7}$.

(b) Checked and absent are independent, so $\mathbb{P}(X) = \frac{1}{7} \cdot \frac{1}{2} = \frac{1}{14} \approx 0.0714$.

(c) Not caught on a given day has probability $1 - \frac{1}{14} = \frac{13}{14}$; over 27 independent days, $\mathbb{P}(\text{never caught}) = \left(\frac{13}{14}\right)^{27} \approx 0.135$.

(d) $\mathbb{P}(\text{caught at least once}) = 1 - \left(\frac{13}{14}\right)^{27} \approx 0.865$.

Drill 5 — Conditional probability from a joint table

$\mathbb{P}(A, B)$	$B = 0$	$B = 1$
$A = 0$	0.1	0.3
$A = 1$	0.2	0.4

Solution. (a) $\mathbb{P}(A=1) = 0.2 + 0.4 = 0.6$. (b) $\mathbb{P}(B=1) = 0.3 + 0.4 = 0.7$.

(c) $\mathbb{P}(A=1 \mid B=1) = \frac{0.4}{0.7} = \frac{4}{7} \approx 0.571$. (d) $\mathbb{P}(B=0 \mid A=0) = \frac{0.1}{0.1 + 0.3} = \frac{1}{4}$.

Drill 6 — Marginalization from a joint distribution

$\mathbb{P}(0, 0) = 0.2, \mathbb{P}(0, 1) = 0.2, \mathbb{P}(1, 0) = 0.1, \mathbb{P}(1, 1) = 0.5$ for (X, Y) .

Solution. (a) $P_X(0) = 0.2 + 0.2 = 0.4, P_X(1) = 0.1 + 0.5 = 0.6$.

(b) $P_Y(0) = 0.2 + 0.1 = 0.3, P_Y(1) = 0.2 + 0.5 = 0.7$.

(c) $\mathbb{P}(Y=1 \mid X=0) = \frac{0.2}{0.4} = 0.5$. (d) $\mathbb{P}(X=1 \mid Y=0) = \frac{0.1}{0.3} = \frac{1}{3}$.

Drill 7 — Conditional probability from a piecewise model

$P_Z(0) = 0.2, P_Z(1) = 0.5, P_Z(2) = 0.3$; $W \mid Z \sim \text{Bernoulli}(Z/4 + 1/4)$.

Solution. The success probability is $\mathbb{P}(W=1 \mid Z=z) = \frac{z}{4} + \frac{1}{4}$: $\frac{1}{4}$ for $z = 0$, $\frac{1}{2}$ for $z = 1$, $\frac{3}{4}$ for $z = 2$.

(a) $\mathbb{P}(W=1 \mid Z=2) = \frac{3}{4}$.

(b) Marginalize: $\mathbb{P}(W=1) = 0.2 \cdot \frac{1}{4} + 0.5 \cdot \frac{1}{2} + 0.3 \cdot \frac{3}{4} = 0.05 + 0.25 + 0.225 = 0.525$.

(c) Bayes: $\mathbb{P}(Z=1 \mid W=1) = \frac{\mathbb{P}(W=1 \mid Z=1)\mathbb{P}(Z=1)}{\mathbb{P}(W=1)} = \frac{0.5 \cdot 0.5}{0.525} = \frac{0.25}{0.525} = \frac{10}{21} \approx 0.476$.

Drill 8 — Marginalization and conditioning with three variables

Table from the worked example:

(a, b, c)	000	001	010	011	100	101	110	111
$p(a, b, c)$	0.1	0.2	0.1	0.1	0.1	0.1	0.1	0.2

Solution. (a) $\mathbb{P}(A=1) = p(100) + p(101) + p(110) + p(111) = 0.1 + 0.1 + 0.1 + 0.2 = 0.5$.

(b) $\mathbb{P}(B=1 \mid A=0) = \frac{p(010) + p(011)}{\mathbb{P}(A=0)} = \frac{0.1 + 0.1}{0.5} = 0.4$.

(c) $\mathbb{P}(C=1 \mid A=1, B=1) = \frac{p(111)}{p(110) + p(111)} = \frac{0.2}{0.1 + 0.2} = \frac{2}{3}$.

(d) $\mathbb{P}(B=0) = p(000) + p(001) + p(100) + p(101) = 0.1 + 0.2 + 0.1 + 0.1 = 0.5$.

Drill 9 — Independence from a model

$Y_B \sim \text{Bernoulli}(2/3)$, $Y_A \mid (Y_B=0) \sim \text{Bernoulli}(1/3)$, $Y_A \mid (Y_B=1) \sim \text{Bernoulli}(1/2)$.

Solution. (a) Use $\mathbb{P}(y_A, y_B) = \mathbb{P}(y_A \mid y_B)\mathbb{P}(y_B)$ with $\mathbb{P}(Y_B=0) = \frac{1}{3}$, $\mathbb{P}(Y_B=1) = \frac{2}{3}$:

$\mathbb{P}(Y_A, Y_B)$	$Y_B = 0$	$Y_B = 1$	
$Y_A = 0$	$\frac{1}{3} \cdot \frac{2}{3} = \frac{2}{9}$	$\frac{2}{3} \cdot \frac{1}{2} = \frac{1}{3}$	$\frac{5}{9}$
$Y_A = 1$	$\frac{1}{3} \cdot \frac{1}{3} = \frac{1}{9}$	$\frac{2}{3} \cdot \frac{1}{2} = \frac{1}{3}$	$\frac{4}{9}$
	$\frac{1}{3}$	$\frac{2}{3}$	1

(Check: $\frac{2}{9} + \frac{1}{3} + \frac{1}{9} + \frac{1}{3} = \frac{2+3+1+3}{9} = 1$.)

(b) $Y_B \sim \text{Bernoulli}(2/3)$ (given). $Y_A \sim \text{Bernoulli}(4/9)$: $\mathbb{P}(Y_A=1) = \frac{1}{9} + \frac{1}{3} = \frac{4}{9}$.

(c) Not independent. Conditionals differ: $\mathbb{P}(Y_A=1 \mid Y_B=0) = \frac{1}{3} \neq \frac{1}{2} = \mathbb{P}(Y_A=1 \mid Y_B=1)$. Product rule: $\mathbb{P}(Y_A=1, Y_B=1) = \frac{1}{3}$ but $\mathbb{P}(Y_A=1)\mathbb{P}(Y_B=1) = \frac{4}{9} \cdot \frac{2}{3} = \frac{8}{27} \neq \frac{9}{27}$.

Drill 10 — Reverse-engineering a joint distribution

$X, Y \in \{0, 1, 2\}$ independent, $P_X = (0.2, 0.5, 0.3)$, $P_Y = (0.4, 0.4, 0.2)$.

Solution. (a) Independence means $\mathbb{P}(x, y) = P_X(x)P_Y(y)$:

	$Y = 0$	$Y = 1$	$Y = 2$	
$X = 0$	0.08	0.08	0.04	0.2
$X = 1$	0.20	0.20	0.10	0.5
$X = 2$	0.12	0.12	0.06	0.3
	0.4	0.4	0.2	1

(b) Add 0.05 to (0, 0) and (1, 1); subtract 0.05 from (0, 1) and (1, 0):

	$Y = 0$	$Y = 1$	$Y = 2$	
$X = 0$	0.13	0.03	0.04	0.2
$X = 1$	0.15	0.25	0.10	0.5
$X = 2$	0.12	0.12	0.06	0.3
	0.4	0.4	0.2	1

Rows: $0.13+0.03+0.04 = 0.2$, $0.15+0.25+0.10 = 0.5$, 0.3 unchanged. Columns: $0.13+0.15+0.12 = 0.4$, $0.03+0.25+0.12 = 0.4$, 0.2 unchanged. Both marginals are unchanged, because each row and column lost exactly as much as it gained.

(c) Not independent: $\mathbb{P}(X=0, Y=0) = 0.13$ but $P_X(0)P_Y(0) = 0.2 \cdot 0.4 = 0.08$.

(d) Two different joint distributions (one independent, one not) have exactly the same marginals, so the marginals do not determine the joint distribution: dependence lives in the joint table and is invisible in the marginals.

Drill 11 — Reading simulation code

```
p = 0.3
samples = np.random.choice([0, 1], p=[1 - p, p], size=1000)
estimate = np.mean(samples)
```

Solution. (a) 1000 iid draws from $\text{Bernoulli}(0.3)$ (each draw is 1 with probability 0.3, else 0).

(b) The sample space of each draw is $\{0, 1\}$.

(c) `estimate` is the fraction of ones, so it approximates $\mathbb{P}(X=1) = 0.3$ (equivalently $\mathbb{E}[X] = 0.3$).

(d) No. The draws are random, so each run gives a slightly different value; the typical deviation from 0.3 is about $\sqrt{0.3 \cdot 0.7/1000} \approx 0.015$.

Drill 12 — Translating code into a joint model

```
y = np.random.choice([0, 1], p=[0.4, 0.6])
if y == 0:
    x = np.random.choice([0, 1], p=[0.8, 0.2])
else:
    x = np.random.choice([0, 1], p=[0.3, 0.7])
```

Solution. (a) $Y \sim \text{Bernoulli}(0.6)$, $X | Y \sim \text{Bernoulli}(0.2 + 0.5Y)$ (i.e. 0.2 if $Y = 0$, 0.7 if $Y = 1$).

(b) $\mathbb{P}(X=1, Y=0) = \mathbb{P}(X=1 | Y=0)\mathbb{P}(Y=0) = 0.2 \cdot 0.4 = 0.08$; $\mathbb{P}(X=1, Y=1) = 0.7 \cdot 0.6 = 0.42$.

(c) $\mathbb{P}(X=1) = 0.08 + 0.42 = 0.50$.

(d) Not independent: $\mathbb{P}(X=1 | Y=0) = 0.2 \neq 0.7 = \mathbb{P}(X=1 | Y=1)$.

Drill 13 — Describing a simulation plan

Check the probabilities from Drill 9 by simulation.

Solution. (a) Each simulated row contains a pair (Y_B, Y_A) : first draw $Y_B \sim \text{Bernoulli}(2/3)$, then draw Y_A from $\text{Bernoulli}(1/3)$ if $Y_B = 0$ and $\text{Bernoulli}(1/2)$ if $Y_B = 1$.

```
n = 100000
yb = np.random.choice([0, 1], size=n, p=[1/3, 2/3])
```

```
ya = np.empty(n, dtype=int)
ya[yb == 0] = np.random.choice([0, 1], size=len(ya[yb == 0]), p=[2/3, 1/3])
ya[yb == 1] = np.random.choice([0, 1], size=len(ya[yb == 1]), p=[1/2, 1/2])
```

- (b) Marginal: the fraction of rows satisfying the event, e.g. `np.mean(ya == 1)` for $\mathbb{P}(Y_A=1)$.
 (c) Conditional: restrict to the rows satisfying the condition, then take the fraction, e.g. `np.mean(ya[yb == 0] == 1)` for $\mathbb{P}(Y_A=1 \mid Y_B=0)$.
 (d) Compare the estimates with the exact values ($\mathbb{P}(Y_A=1) = \frac{4}{9} \approx 0.444$, $\mathbb{P}(Y_A=1 \mid Y_B=0) = \frac{1}{3}$, $\mathbb{P}(Y_A=1 \mid Y_B=1) = \frac{1}{2}$, $\mathbb{P}(Y_A=1, Y_B=1) = \frac{1}{3}$). They should agree up to sampling error of roughly $1/\sqrt{n}$; a larger n shrinks the gap. (Ran it with $n = 2$ million: 0.444, 0.333, 0.500, 0.333.)

Drill 14 — Reading NumPy array code

```
x = np.array([4, 7, 7, 1, 3])
```

- Solution.** (a) `x[2] = 7` (indexing starts at 0: $x[0] = 4$, $x[1] = 7$, $x[2] = 7$).
 (b) `x[x >= 5] = array([7, 7])` (the entries that are at least 5).
 (c) `np.mean(x == 7)`: the boolean array is `[F, T, T, F, F]`, $\text{mean} = 2/5 = 0.4$.
 (d) `len(x[x > 3])`: entries greater than 3 are 4, 7, 7, so the length is 3.

Drill 15 — Marginal, joint, conditional probability from arrays

- Solution.** (a) `np.mean(x == 1)`
 (b) `np.mean((x == 1) & (y == 0))`
 (c) `np.mean(y[x == 1] == 0)` (restrict to rows with $x = 1$, then take the fraction with $y = 0$).
 (d) $\mathbb{P}(Y=0 \mid X=1) = \frac{\mathbb{P}(X=1, Y=0)}{\mathbb{P}(X=1)} \approx \frac{0.18}{0.45} = 0.4$.

Drill 16 — Uniform probabilities

$Y \sim \text{Uniform}(0, 10)$, density $f = \frac{1}{10}$ on $[0, 10]$.

- Solution.** (a) $\mathbb{P}(2 < Y < 5) = \frac{5-2}{10} = 0.3$. (b) $\mathbb{P}(Y > 7) = \frac{10-7}{10} = 0.3$. (c) $\mathbb{P}(Y = 4) = 0$.
 (d) A single point has zero length, so zero area under the density. This does not mean Y can't be near 4: intervals around 4 have positive probability (e.g. $\mathbb{P}(3.9 < Y < 4.1) = 0.02$), and $\mathbb{P}(Y = 4) = 0$ is just the limit as the interval shrinks to a point.

Drill 17 — Density is not probability

$Y \sim \text{Uniform}(0, 0.2)$.

- Solution.** (a) $f(y) = \frac{1}{0.2} = 5$ on $[0, 0.2]$ (and 0 elsewhere).
 (b) A density is probability *per unit length*, not a probability. Only the area has to be at most 1: here $5 \times 0.2 = 1$.
 (c) $\mathbb{P}(0.05 < Y < 0.10) = 5 \times 0.05 = 0.25$.

Drill 18 — Conditioning on an interval

$Y \sim \text{Uniform}(0, 1)$.

Solution. (a) $\mathbb{P}(Y < 0.3) = 0.3$.

(b) $\mathbb{P}(Y < 0.3 \mid Y < 0.5) = \frac{\mathbb{P}(Y < 0.3)}{\mathbb{P}(Y < 0.5)} = \frac{0.3}{0.5} = 0.6$ (since $\{Y < 0.3\} \subset \{Y < 0.5\}$).

(c) $\mathbb{P}(Y > 0.7 \mid Y > 0.4) = \frac{\mathbb{P}(Y > 0.7)}{\mathbb{P}(Y > 0.4)} = \frac{0.3}{0.6} = 0.5$.

Drill 19 — Mixing a Bernoulli and a Uniform

$X \sim \text{Bernoulli}(0.3)$ (rain), $Y \mid (X=0) \sim \text{Uniform}(0, 5)$, $Y \mid (X=1) \sim \text{Uniform}(0, 20)$.

Solution. (a) $\mathbb{P}(Y < 3 \mid X=0) = \frac{3}{5} = 0.6$; $\mathbb{P}(Y < 3 \mid X=1) = \frac{3}{20} = 0.15$.

(b) Marginalize X out: $\mathbb{P}(Y < 3) = 0.7(0.6) + 0.3(0.15) = 0.42 + 0.045 = 0.465$.

(c) Bayes: $\mathbb{P}(X=1 \mid Y < 3) = \frac{\mathbb{P}(Y < 3 \mid X=1)\mathbb{P}(X=1)}{\mathbb{P}(Y < 3)} = \frac{0.15 \cdot 0.3}{0.465} = \frac{0.045}{0.465} = \frac{3}{31} \approx 0.097$.

(d) A short delay is much more likely on a dry day (0.6) than a rainy one (0.15), so seeing a short delay makes rain less likely than the prior 0.3.

Drill 20 — Simulating a mixed model

Solution.

```
N = 200_000
x = np.random.choice([0, 1], size=N, p=[0.7, 0.3])      # X ~ Bernoulli(0.3)
y = np.empty(N)
y[x == 0] = np.random.uniform(0, 5, len(y[x == 0]))     # Y | X=0 ~ Uniform(0,5)
y[x == 1] = np.random.uniform(0, 20, len(y[x == 1]))     # Y | X=1 ~ Uniform(0,20)

print(np.mean(y < 3))                                   # estimates P(Y < 3), exact 0.465
print(np.mean(x[y < 3] == 1))                           # estimates P(X=1 | Y<3), exact 3/31 = 0.0968
```

The first print should be close to 0.465 (Drill 19(b)); the second restricts to the samples with $y < 3$ and averages the indicator $x = 1$, so it should be close to 0.097 (Drill 19(c)). (Simulated with 2 million draws: 0.465 and 0.097.)

Drill 21 — Binomial probabilities

$Y \sim \text{Binomial}(4, q)$.

Solution. (a) $S_Y = \{0, 1, 2, 3, 4\}$ (number of successes in 4 trials).

(b) $\mathbb{P}(Y = 0) = (1 - q)^4$ (all four fail).

(c) $\mathbb{P}(Y = 1) = 4q(1 - q)^3$ (four positions for the single success, each sequence has probability $q(1 - q)^3$).

(d) $\mathbb{P}(Y \geq 1) = 1 - \mathbb{P}(Y = 0) = 1 - (1 - q)^4$ (the complement is the simplest route).

Drill 22 — Counting configurations

$Y \sim \text{Binomial}(5, q)$.

Solution. (a) $\binom{5}{2} = \frac{5 \cdot 4}{2 \cdot 1} = 10$ sequences of 5 flips have exactly two ones.

(b) One particular sequence, e.g. $(1, 1, 0, 0, 0)$: $q \cdot q \cdot (1 - q)^3 = q^2(1 - q)^3$ (independence).

(c) $\mathbb{P}(Y = 2) = \binom{5}{2} q^2(1 - q)^3 = 10 q^2(1 - q)^3$.

(d) $q^2(1 - q)^3$ is the probability of only *one* arrangement of the two ones; $Y = 2$ happens for any of the 10 arrangements, and they are disjoint, so their probabilities add.

2 Unit 2 drills**Computing conditional averages**

Data $\{(1, 2), (1, 2), (3, 1), (1, 4), (3, 3), (2, 2), (1, 5)\}$ of pairs (Y_1, Y_2) .

Solution. (a) $\mathbb{E}[Y_1]$ is estimated by the average of *all seven* Y_1 values: $\frac{1 + 1 + 3 + 1 + 3 + 2 + 1}{7} = \frac{12}{7} \approx 1.71$.

(b) $\mathbb{E}[Y_1 \mid Y_2=2]$: keep the rows with $Y_2 = 2$: $(1, 2), (1, 2), (2, 2)$; average their Y_1 : $\frac{1 + 1 + 2}{3} = \frac{4}{3}$.

(c) $\mathbb{E}[Y_2 \mid Y_1=1]$: rows with $Y_1 = 1$: $(1, 2), (1, 2), (1, 4), (1, 5)$; average their Y_2 : $\frac{2 + 2 + 4 + 5}{4} = \frac{13}{4} = 3.25$.

(d) $\mathbb{E}[Y_2 \mid Y_1 > 1]$: rows with $Y_1 > 1$: $(3, 1), (3, 3), (2, 2)$; average their Y_2 : $\frac{1 + 3 + 2}{3} = 2$.

Conditional means from a joint table

	$X = 0$	$X = 1$	$X = 2$
$Y = 0$	0.10	0.20	0.10
$Y = 1$	0.15	0.15	0.30

Solution. Column sums (marginal of X): $\mathbb{P}(X=0) = 0.25$, $\mathbb{P}(X=1) = 0.35$, $\mathbb{P}(X=2) = 0.40$.

$$\mathbb{E}[X] = 0(0.25) + 1(0.35) + 2(0.40) = 1.15.$$

Row $Y = 0$ sums to 0.4: $\mathbb{E}[X \mid Y=0] = \frac{0(0.10) + 1(0.20) + 2(0.10)}{0.4} = \frac{0.4}{0.4} = 1$.

Row $Y = 1$ sums to 0.6: $\mathbb{E}[X \mid Y=1] = \frac{0(0.15) + 1(0.15) + 2(0.30)}{0.6} = \frac{0.75}{0.6} = 1.25$.

Check with the tower property: $0.4(1) + 0.6(1.25) = 1.15 = \mathbb{E}[X]$. ✓

Expectation and variance from a table

$\mathbb{P}(Y=0) = 0.2$, $\mathbb{P}(Y=1) = 0.5$, $\mathbb{P}(Y=2) = 0.3$.

Solution. $\mathbb{E}[Y] = 0(0.2) + 1(0.5) + 2(0.3) = 1.1$. $\mathbb{E}[Y^2] = 0(0.2) + 1(0.5) + 4(0.3) = 1.7$.
 $\text{var}(Y) = \mathbb{E}[Y^2] - (\mathbb{E}[Y])^2 = 1.7 - 1.21 = 0.49$ (so $\sigma = 0.7$).

Weighted conditional means

$\mathbb{P}(X=0) = 0.7$, $\mathbb{P}(X=1) = 0.3$, $\mathbb{E}[Y \mid X=0] = 10$, $\mathbb{E}[Y \mid X=1] = 20$.

Solution. Tower property: $\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y \mid X]] = 0.7(10) + 0.3(20) = 7 + 6 = 13$.

Law of total variance from conditional moments

Same setup plus $\text{var}(Y \mid X=0) = 4$, $\text{var}(Y \mid X=1) = 9$.

Solution. Within-group: $\mathbb{E}[\text{var}(Y \mid X)] = 0.7(4) + 0.3(9) = 2.8 + 2.7 = 5.5$.

Between-group: $\text{var}(\mathbb{E}[Y \mid X]) = 0.7(10 - 13)^2 + 0.3(20 - 13)^2 = 0.7(9) + 0.3(49) = 6.3 + 14.7 = 21$
(using $\mathbb{E}[Y] = 13$ from the previous drill).

Total: $\text{var}(Y) = 5.5 + 21 = 26.5$.

Sum of Bernoulli variables

$Y = \sum_{i=1}^{12} X_i$, X_i iid Bernoulli(0.25).

Solution. $\mathbb{E}[Y] = 12(0.25) = 3$. $\text{var}(Y) = 12(0.25)(0.75) = 2.25$, so $\sigma = 1.5$ (variances add for independent X_i).

CV = $\sigma/\mu = 1.5/3 = 0.5$. (Check with the formula $\sqrt{(1-q)/(qN)} = \sqrt{0.75/3} = 0.5$.)

Normal probabilities

$Y \sim \text{Normal}(10, 9)$.

Solution. The variance is 9, so $\sigma = 3$.

(a) $Z = \frac{Y - 10}{3} \sim \text{Normal}(0, 1)$.

(b) $13 = 10 + 3 = \mu + \sigma$, so $\mathbb{P}(Y > 13) = \mathbb{P}(Z > 1) \approx \frac{100 - 68}{2}\% = 16\%$.

(c) $7 = \mu - \sigma$ and $13 = \mu + \sigma$, so $\mathbb{P}(7 < Y < 13) \approx 68\%$.

Linear transformation

$X \sim \text{Normal}(5, 9)$, $Y = -2X + 7$.

Solution. (a) $\mathbb{E}[Y] = -2(5) + 7 = -3$; $\text{var}(Y) = (-2)^2 \cdot 9 = 36$. So $Y \sim \text{Normal}(-3, 36)$, with $\sigma_Y = 6$.

(b) Standardize: $\frac{Y + 3}{6}$. Since $3 = -3 + 6 = \mu_Y + \sigma_Y$, $\mathbb{P}(Y > 3) \approx 16\%$.

(c) $\mathbb{P}(Y > -3) = \frac{1}{2}$: -3 is the mean of Y and a Normal distribution is symmetric about its mean.

Sum of Normal random variables

$X_1 \sim \text{Normal}(2, 4)$, $X_2 \sim \text{Normal}(-1, 9)$ independent, $S = 3X_1 - 2X_2$.

Solution. (a) $\mathbb{E}[S] = 3(2) - 2(-1) = 8$. $\text{var}(S) = 3^2(4) + (-2)^2(9) = 36 + 36 = 72$ (independence, so variances add with squared coefficients). $S \sim \text{Normal}(8, 72)$, $\sigma_S = \sqrt{72} = 6\sqrt{2} \approx 8.49$.

(b) $\frac{S - 8}{\sqrt{72}} \sim \text{Normal}(0, 1)$.

(c) 0 and 16 are each 8 away from the mean, and $8/8.49 \approx 0.94$ standard deviations. So the interval is a bit narrower than $\mu \pm 1\sigma$, and the estimate is **a bit under** 68%, roughly 65%. (The exact value is 0.654; on an exam, “about 65–68%, just below the $\pm 1\sigma$ rule” is the expected level of precision.)

Binary predictor means

$X \in \{0, 1\}$, $Y \mid X \sim \text{Normal}(\beta_0 + \beta_1 X, \sigma^2)$.

Solution. $\mathbb{E}[Y \mid X=0] = \beta_0$; $\mathbb{E}[Y \mid X=1] = \beta_0 + \beta_1$; difference = β_1 .

Why the sample version is natural: a conditional expectation can be estimated by the sample average over the rows satisfying the condition. So $\bar{Y}_{X=1} \approx \mathbb{E}[Y \mid X=1]$ and $\bar{Y}_{X=0} \approx \mathbb{E}[Y \mid X=0]$, and their difference estimates β_1 ; in code, `y[x==1].mean() - y[x==0].mean()`.

Covariance, correlation, and shrinkage

$\text{var}(X) = 4$, $\text{var}(Y) = 25$, $\text{cov}(X, Y) = 8$.

Solution. (a) $\beta_1 = \frac{\text{cov}(X, Y)}{\text{var}(X)} = \frac{8}{4} = 2$ (from $\text{cov}(X, Y) = \beta_1 \sigma_X^2$).

(b) $\rho = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{8}{2 \cdot 5} = 0.8$. (Check: $\beta_1 \sigma_X / \sigma_Y = 2 \cdot 2/5 = 0.8$.)

(c) $\mathbb{E}[Z_Y \mid \bar{Z}_X=2] = \rho \cdot 2 = 1.6$ standard deviations above the mean of Y : less extreme than X 's 2 (regression to the mean, since $|\rho| < 1$).

Linear regression model parameters

$\hat{\beta}_1 = \frac{1}{2}$, $\hat{\sigma}_\epsilon = 1$, $\sigma_X^2 = 4$. Estimate R^2 .

Solution. $\sigma_Y^2 = \beta_1^2 \sigma_X^2 + \sigma_\epsilon^2 = \frac{1}{4} \cdot 4 + 1 = 2$.

$R^2 = 1 - \frac{\sigma_\epsilon^2}{\sigma_Y^2} = 1 - \frac{1}{2} = 0.5$. (Equivalently $\frac{\beta_1^2 \sigma_X^2}{\sigma_Y^2} = \frac{1}{2}$: half of the variance of Y is explained by X .)

Correlation and units

$\sigma_X = 10$, $\sigma_Y = 4$, $\text{cov}(X, Y) = 20$.

Solution. (a) $\rho = \frac{20}{10 \cdot 4} = 0.5$.

(b) Slope $\beta_1 = \frac{\text{cov}(X, Y)}{\sigma_X^2} = \frac{20}{100} = 0.2$. (Check: $\rho \sigma_Y / \sigma_X = 0.5 \cdot 4/10 = 0.2$.)

(c) $\text{cov}(X, Y)$ has units (X-units) $\times$ (Y-units), and $\sigma_X \sigma_Y$ has the same units, so they cancel in ρ : it is unitless (it is the slope after standardizing both variables). The slope $\beta_1 = \text{cov} / \sigma_X^2$ has units (Y-units)/(X-units): the change in Y per one-unit change in X . For example, measuring X in meters instead of centimeters multiplies β_1 by 100 but leaves ρ unchanged.